

Macro HW1

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Question 1. Every entry is a percent deviation from trend.

Variable	% Standard Deviation	Correlation with Output
Output	1.70	1.00
Consumption	1.60	0.81
Investment	6.45	0.63
Gov't Consumption Expenditure	1.01	0.22
Gov't Investment Expenditure	3.20	-0.06
Exports	3.70	0.76
Imports	4.72	0.82
Net Exports	0.69	-0.30
Hours Worked	2.16	0.89
Labour Productivity	1.01	-0.23
Capital	0.87	-0.15
Capital Productivity	2.02	0.90
Nominal Exchange Rate	3.86	-0.12
Price	1.32	0.25

$$\text{corr}(\text{hours worked, labour productivity}) = -0.65$$

Question 2.

(a)



Figure 1. GDP per capita: data and HP trend (top), cycle (bottom).

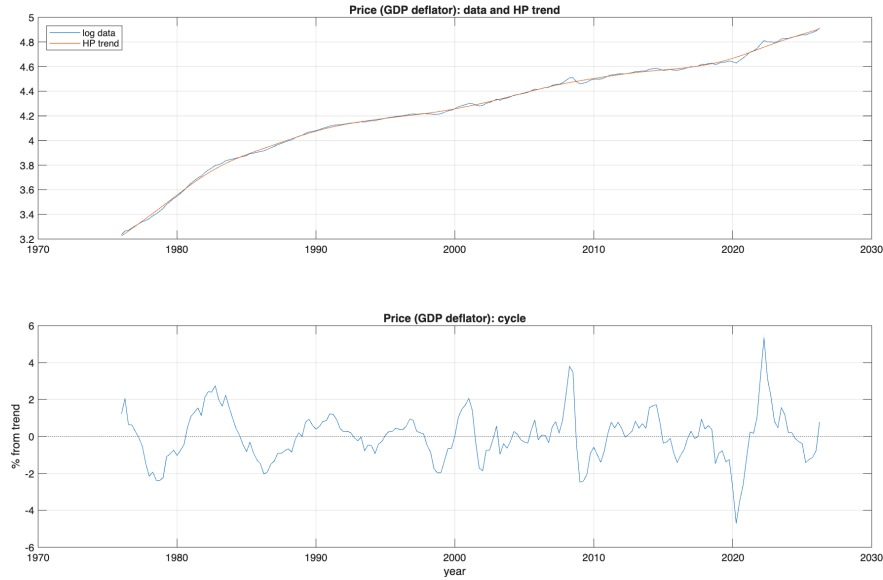


Figure 2. Price (GDP deflator): data and HP trend (top), cycle (bottom).

(b) GDP per capita peaks in 2008Q3, so that is where the recession starts. Output falls for three quarters, drops below trend in 2009Q1 and bottoms in 2009Q2. The recovery starts there, and output is back at trend in 2011Q3.

(c)

Episode	Peak	Trough	Depth (%)	Duration (quarters)
Early 1980s	1981Q2	1982Q4	-5.44	6
Early 1990s	1989Q1	1992Q2	-2.29	13
Great Recession	2008Q3	2009Q2	-3.42	3

Depth is the lowest point of the cycle and duration runs peak to trough. The Great Recession was about 1.5 times as deep as the early 1990s, roughly 60% as deep as the early 1980s, and at three quarters the shortest of the three. The early 1990s was the slowest: output stayed below trend for 14 quarters, against 11 in the early 1980s and 10 in the Great Recession.

The 1990s start is a close call. The cycle also peaks at 1988Q2 (+2.56%), a hair above 1989Q1 (+2.55%). I use 1989Q1 because it starts the sustained decline; counting from 1988Q2 would make the episode 16 quarters instead of 13.

Question 3. A collapse in demand. Output and prices fell together over 2007Q1–2011Q4 (correlation +0.83), with the price cycle bottoming at -2.5% in 2009Q1 and output at -3.4% in 2009Q2. A supply shock would have pulled the two apart: over 1979Q1–1983Q4, the oil-shock years, the correlation is -0.61, with output 5.4% below trend while prices ran 2.8% above it. The collapse itself came from the financial crisis: credit tightened, households lost wealth, firms cut investment, and U.S. demand for Canadian exports fell.

Question 4.

Variable	Fall, 2019Q4–2020Q2	Cycle low	Back at 2019Q4 level
Output (GDP p.c.)	−13.1%	−12.6% (2020Q2)	2021Q4
Price (deflator)	−1.7%	−4.7% (2020Q2)	2020Q3
Hours worked	−21.9%	−21.9% (2020Q2)	2023Q1

The falls are percent changes in the population-deflated level from 2019Q4 to 2020Q2. Output dropped 13.1%, and the trough was 3.7 times deeper than the Great Recession’s, but output was back at its pre-pandemic level by 2021Q4. Hours fell 21.9%, so output per hour rose, and hours were the last to recover, not until 2023Q1. Prices dipped only 1.7%.